Self-Healing Autonomous Mesh For Resilient naval communication

Project report submitted
In partial fulfillment of the requirements
For the award of the Degree of

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In
Electronics and Telecommunication Engineering

by

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Declaration

We, the students Adarsh Ramgirwar(2022BEC155), Saikrishna Vaijwade(2022BEC123), Avnish Joshi(2022BEC091) hereby declare that this project work titled "Self-Healing Autonomous Mesh for Resilient Naval Communication" is carried out by us in the Department of Electronics and Telecommunication Engineering of Shri Guru Gobind Singhji Institute of Engineering &Technology, Vishnupuri, Nanded. The work is original and has not been submitted earlier whole or in part for the award of any degree/diploma at this or any other Institution / University.

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Certificate

This is to certify that the project titled "Self-Healing Autonomous Mesh for Resilient Naval Communication", submitted by student Adarsh Ramgirwar(2022BEC155), Saikrishna Vaijwade(2022BEC123), Avnish Joshi(2022BEC091) in partial fulfillment of the requirements for the 8th Semester Major Project of **Bachelor of Technology in Electronics and Telecommunication Engineering**, SGGSIE&T, Nanded. The work is comprehensive, complete and fit for final evaluation.

Prof. A.V Nandedkar
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Mr. Sunil Jatti
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Abstract

Communication breakdown is one of the most critical challenges faced during disasters, maritime emergencies, and conflict situations, especially in remote or offshore environments where conventional network infrastructure is unavailable. This project presents a **Mesh Networking-Based Communication System for Fishermen and Disaster Management**, designed to enable reliable, decentralized communication without dependence on centralized networks such as cellular or internet services.

The system leverages low-cost wireless devices (e.g., ESP-based nodes) configured in a mesh topology, where each node can transmit, receive, and relay messages across the network. This architecture ensures that even if direct communication between two nodes is not possible, messages can propagate through intermediate nodes (e.g., nearby boats) until they reach the intended destination or base station. Each device is assigned a logical identity, enabling targeted communication, status updates, and emergency signaling.

The proposed solution allows fishermen to share critical information such as distress signals, location updates, and safety alerts with nearby vessels and coastal authorities. In disaster scenarios, where infrastructure is damaged or unavailable, the mesh network dynamically adapts to node availability, ensuring continued communication. Additionally, in high-risk or conflict environments, the system enables secure, short-range communication between units, enhancing coordination and situational awareness.

By eliminating reliance on centralized infrastructure and enabling self-healing communication pathways, this system significantly improves resilience, safety, and response efficiency in challenging environments. The solution offers a scalable, cost-effective, and robust approach to emergency communication for maritime and disaster management applications.

List of Figures

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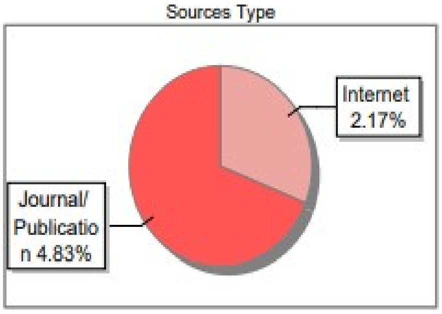

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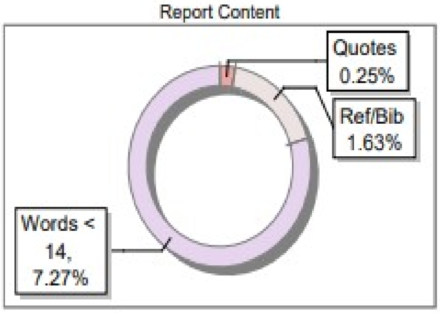

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Chapter 1: Introduction

Over the last few years, networks of communication has played a central role in numerous technological advancements for use in various application areas such as disaster management, military applications, remote control, monitoring, and widespread IoT systems. Standard wired communication networks generally use centralized architectures (routers, base stations, or servers) for the efficient management of data traffic. Though they offer high efficiency under the best conditions, they have a serious drawback; their failure. A breakdown in a central node or communication line will affect the overall system. Decentralized networking architectures (e.g. Mesh networks) have gained interest over the last few years due to their ability to offer robustness in the face of node and network failures. A mesh network is defined as a topology where any two nodes can communicate with one another, directly or indirectly through intermediate nodes; and hence they can be highly resilient as a failed node can always be bypassed. Nevertheless, traditional mesh networks usually do not offer sophisticated solutions to automate failures and adapt routing dynamically, that is required for mission critical environments.

Self-Healing Autonomous Mesh (SHAM) system is a decentralized, self-organizing network system which integrates intelligence and dynamism into traditional mesh networking for achieving self-healing properties for its system. Each node in this architecture is independent and cooperative to each other for effective network communication and automatically detects, adapts and responds to changes in the topology and recovers from failure without any external human intervention.

In the SHAM architecture, nodes (using devices like ESP32) will be programmed with routing algorithms and detection schemes. The nodes in the system will exchange broadcast messages periodically which will also be known as "HELLO" message, to determine and discover reachable neighbors. Using the routing information, each node maintains its routing table, which can list the network topology, information about reachable neighbors and best paths between the nodes in the network topology.

The system will then determine best paths through a number of sophisticated routing algorithms, for example Dijkstra's shortest path algorithm, through the current network state dynamically. Unlike static routing systems where paths are hardwired, SHAM will determine new routes for the packet in the current state dynamically and hence provide efficient communication path with minimal latency. Other metrics like Received Signal Strength Indicator (RSSI), latency, congestion and many more may also be incorporated to refine dynamic path calculation to find more appropriate routes for effective data transmission.

Furthermore, SHAM system will perform dynamic self-healing. In real world scenario it may be possible that certain nodes in the system might have failed due to hardware problems or loss of power etc. In order to compensate these issues, SHAM utilizes timeout and ACKs. Timeout will enable the detection of node failure when a node has not responded within a specified duration. Nodes that do not respond will be automatically discarded and the system recalculates the new route for the communication by utilizing the other active node in the mesh network. This self-healing characteristic is crucial in mission critical applications where reliable communication is required.

Centralized control is eliminated making the system to be more resistant against system failures, enabling to provide scalability and extensibility. Nodes can join or leave the network dynamically, as new nodes will detect the network through discovery messages. This makes it compatible with mobile and temporary network configurations. Multi-hop communication can also be achieved, where messages can be transmitted over long distance through other intermediate nodes which is a very useful attribute when nodes are not within direct communication range and have physical obstacles between them.

However, complexities, power consumptions and security issues may be presented during the implementation of SHAM system; they may be solved by employing effective routing techniques, smart coding protocols and secure encryptions.

Self-Healing Autonomous Mesh (SHAM) system presents a unique and an effective approach to achieve distributed and reliable network communication. The integration of dynamic routing, self-healing properties, and autonomous network operation provides an innovative solution to the modern communication challenges. The aim of this project is to demonstrate the design and implementation of such system utilizing theESP32 device as nodes of the network and simulated the communication between nodes.

Chapter 2: Evolution

The design of Self-Healing Autonomous Mesh (SHAM) system is an evolution from communication network development which progresses toward increasing the network reliability, scalability and fault tolerance. Throughout the history, networking paradigms moved from centralized architectures to distributed, intelligent systems which can operate under dynamically changing network environment. The section discusses the historical evolution of the technologies that paved way for the design of SHAM system.

2.1 Traditional centralized networks

Early communication systems use the centralized network approach. In this, only a single server/control node handles the responsibility of managing, routing and operating the whole network and each device is reliant on the central node. Although simple and easy to manage and implement but there lies one serious issue, 'Single Point of Failure'. If the central node fails then the entire network becomes unusable. Centralized networks are not useful for mission critical applications which require utmost reliability.

2.2 Distributed networks

To compensate the issues that centralized networks possess, distributed networks were designed. In this approach, the task is distributed across several nodes and a node no longer depends entirely on a single server to perform its tasks. These systems improve reliability and fault tolerance. However these do not achieve effective dynamic routing and automatic failure recovery in dynamic environment.

2.3 Mesh networks

Mesh networks were another improvement over distributed networks, since the nodes can communicate with more than one node at a time. In mesh networks, the data can reach to any end through several different intermediate nodes. This type of topology improve network coverage and reliability. Mesh networks did lack of intelligent dynamic routing capabilities and fault recovery for dynamic topology.

2.4 Mobile Ad hoc Networks (MANETs)

Mobile Ad hoc Networks (MANETs) offer networks that are self-configuring and allow nodes to dynamically create connections without any need for a preexisting infrastructure. MANETs typically utilize routing protocols such as AODV (Ad hoc On-Demand Distance Vector) and DSR (Dynamic Source Routing) for establishing a path between nodes. MANETs are well-suited for mobile environments. MANETs has issues like high routing overhead and latency, poor scalability etc.

2.5 IoT and wireless embedded systems

ESP32 is a low power wireless embedded system that forms a crucial part of IoT based networks, used for monitoring and transmitting the data in real time wireless sensor networks. However these systems generally rely on the centralized server or gateway, thereby bringing back the Single point of failure. These are also not self-contained.

2.6 Need of self-healing, autonomous networks

With increasing use of communication networks in various field such as disaster management, military communication and remote monitoring etc. It became imperative to have an autonomous network which can also repair itself, recover from failure without human intervention and can perform its function even in hostile environments.

2.7 Emergence of SHAM (Self-Healing Autonomous Mesh)

SHAM is an integration of the all previously described concepts in networking technology to provide a high performant, fully autonomous and self-healing network. It integrates distributed network, mesh network, mobile adhoc networks and low power IoT devices. It makes use of intelligent routing algorithms like Dijkstra to find a path, HELLO messages to establish node discovery and timeout mechanism for detecting a node failure to identify the broken link. The SHAM is a self-contained system that can dynamically adapt itself to varying network topologies and failures. Unlike existing systems it does not requires a centralized control node.

Chapter 3: System Design And Architecture

The Self-Healing Autonomous Mesh (SHAM) system provides an intelligent and decentralized communication mechanism to ensure robust data transmission over highly dynamic and failure-prone networks. It integrates autonomous node operation, adaptive routing, and self-healing abilities. Each node functions as both a communication endpoint and a router to facilitate efficient multi-hop data routing, eliminating the dependency on any central infrastructure.

3.1 System Overview

The SHAM system consists of multiple interconnected wireless nodes (either actual ESP32 devices or simulated nodes) connected via a mesh topology. Each node in the network is designed to discover neighboring nodes, manage its routing information, transmit and receive data packets, and detect any node failures. The system operates in a fully distributed manner, with all nodes contributing to maintaining reliable and efficient network communication.The system architecture is organized into three distinct layers: Node Layer, Network Layer and Application Layer.

3.2 Node Architecture

Each node within the SHAM system is structured as an autonomous entity equipped with various functional modules that enable it to perform its designated tasks. The key components that make up a node are: communication module, routing module, failure detection mechanism and a data handler.Communication Module: The node uses the Wi-Fi functionality of an ESP32 to send and receive messages. The module supports broadcasting for neighbor discovery and unicast communication for actual data packet forwarding.

Routing Module: This module is responsible for maintaining the routing table, which stores information about the network topology and the most cost-effective path to reach different destinations. It employs a shortest path algorithm like Dijkstra's to determine the best route for data packets.

Failure Detection Module: The module keeps track of active nodes by using timeout mechanisms and missing acknowledgment signals from nodes. When a node becomes unresponsive, this module flags it as inactive and triggers the removal of its entry from the routing table.

Data Buffer: A buffer where data packets are stored both when incoming and outgoing, before they are processed by the node or forwarded to the next node in the path.

3.3 Network Architecture (Mesh Topology)

The SHAM system is built using a mesh network topology. In a mesh network, nodes have direct connections to each other, making it possible for data to take multiple routes and paths to reach its destination. The key characteristics of the SHAM mesh network topology include:

Nodes are directly connected if within their wireless range

Data is forwarded through intermediate nodes if its destination is not within range (multi-hop)

Multiple paths between any two nodes exist, allowing for greater reliability

This decentralized topology makes the system independent of a central controller and also ensures a degree of fault tolerance since there are alternative routes if a particular node or link fails.

3.4 Routing Mechanism

The SHAM system uses a dynamic routing mechanism which allows for real-time adaptation of routes to accommodate network topology changes, and other conditions. Every node has a routing table to store the list of destinations that it can reach, along with a cost metric that is associated with each path.

The shortest path is computed using Dijkstra's algorithm, with the following cost metrics:

Hop Count: The number of nodes that a data packet needs to pass through from the source node to the destination node.

Signal strength of the connection between two nodes (RSSI) (Optional advanced feature).Transmission latency between two nodes (Optional advanced feature).

Whenever a topology change occurs, due to node additions or failures in the network, the routing table will update dynamically. Following that update, all relevant nodes will recompute the shortest path accordingly to ensure optimal data transmission routes are being used.

3.5 Node Discovery Mechanism

Node discovery in the SHAM system is carried out through periodic broadcasting. All nodes in the network will periodically broadcast a HELLO message in an attempt to find neighbors. If neighboring nodes discover a HELLO message from a previously unknown node, they will reply to the HELLO message with an ACK. Once an ACK is received, the sender will then add the new node to its list of neighbors and update the routing table accordingly to include information about paths that can go through this new node.

3.6 Failure Detection and Self-Healing

One of the most significant features of the SHAM system is its self-healing capability. If a particular node is found to be unresponsive after a certain period, (based on a predetermined timeout) it will be marked as inactive. It will then be removed from the routing table of its neighbors. Following that, the system's shortest path algorithm is used again to find a replacement route which doesn't utilize the failed node, enabling uninterrupted communication between nodes.

3.7 Data Transmission Flow

Data transmission within the SHAM system can be broken down into the following sequence of events:

- A data packet is created by a source node.
- The routing module uses the system's current topology to calculate the shortest path.
- The data packet is transmitted to the first intermediate node on the calculated path.
- Each intermediate node then forwards the data packet along the determined path until it reaches its destination.

- Once the destination node receives the data, it sends an ACK packet back to the source node to confirm delivery.
- If any part of the transmission is not confirmed or if a failure is detected at any step of the communication flow, the system automatically invokes the self-healing mechanism and finds an alternative route.

3.8 Communication Protocol Design

The SHAM system utilizes lightweight messages to optimize the efficiency of the communication process between nodes. The primary message types utilized include:

HELLO Message: Transmitted periodically for discovering and announcing the presence of a node in the network.

DATA Packet: Actual data to be transferred between nodes.

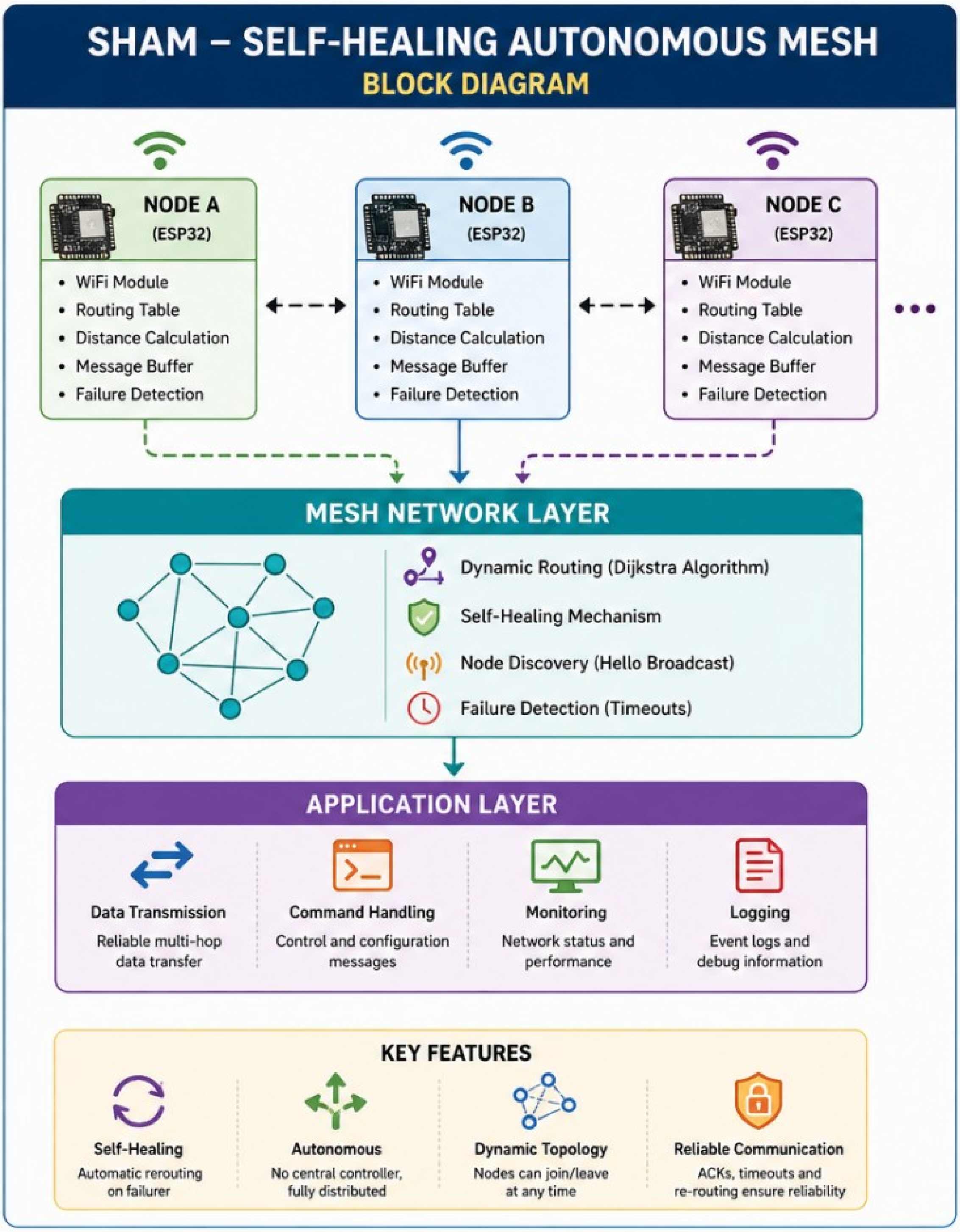
ACK Packet: Used by destination or intermediate nodes to confirm receipt of a packet.

ERROR Message: Used by nodes to inform other nodes about detected failures or issues in the network.

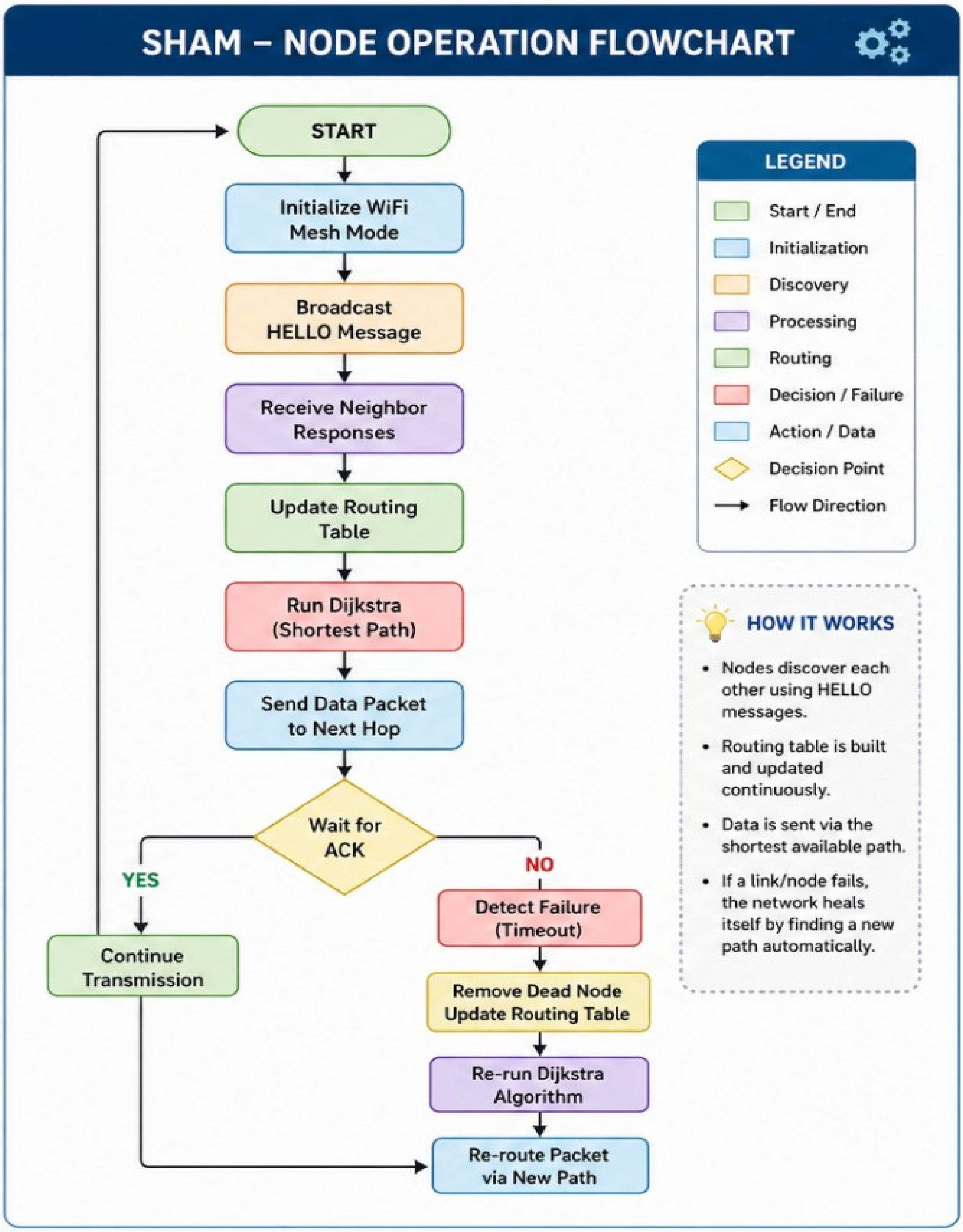
These message types have been designed with minimal overhead, allowing for faster and more reliable data exchange.

3.9 Scalability and Performance Considerations

The SHAM system is designed to be scalable, meaning it can handle an increasing number of nodes and expanding network sizes. However, certain factors need to be taken into consideration as the network grows in size. As the number of nodes increases, the routing table will also grow significantly. This increases the memory requirements of each node and can also lead to increased control message overhead in larger networks. Power consumption in each node must also be efficiently managed. Techniques such as reducing the rate at which HELLO messages are sent and implementing selective routing updates can enhance the performance of larger networks.



Self Healing Autonomous Mesh Block Diagram



SHAM Node Operation Flowchart

Chapter 4: Implementation and working

The Self-Healing Autonomous Mesh (SHAM) system has been implemented using ESP32 microcontrollers which exploit their native Wi-Fi functionality to form an ad-hoc mesh network. The system comprises multiple independent ESP32 nodes where each node is responsible for a set of tasks including discovering neighbors, maintaining routing information, forwarding data, and detecting failures. It is primarily designed to offer a reliable and self-healing communication solution that is not reliant on any central component.

4.1 Hardware Components

The hardware implementation of SHAM has been implemented using the following hardware:

* ESP32 Microcontroller:

This component serves as the primary computing and communication module of each node. It comes equipped with inherent Wi-Fi capabilities, sufficient processing power and memory for the implementation of routing logic.

* Power Supply:

Each ESP32 node is powered either via USB or battery module depending on the specific context of deployment.

4.2 Software Environment

The SHAM system software implementation consists of:

* Arduino IDE / ESP-IDF to develop the software for the ESP32
* C/C++ programming language to code the system's functions
* Wi-Fi / ESP-NOW / Mesh libraries for handling network communication

All nodes will run the same firmware allowing for similar network operations.

4.3 Node Initialization

On powering the ESP32 node, the following tasks will be performed:

* Initialization of Wi-Fi in the appropriate mode (either station mode or mesh mode)
* Assignment of unique ID to the node (using its MAC address)
* Initial creation of routing table and neighbor list
* Starting periodic timers to periodically send out HELLO messages.

4.4 Node Discovery Mechanism

The discovery of neighboring nodes occurs via periodic broadcasts. The workflow is as follows:

* Each ESP32 node broadcasts HELLO messages at a fixed time interval to discover its neighbors.
* Neighbors detect the message based on Wi-Fi reception.
* Neighbor nodes send a response message in the form of an acknowledgement (ACK).
* The original sender can then use this ACK to update its neighbor list.

4.5 Routing Table Management

Each node will maintain its own routing table to decide the optimal path to any other node on the mesh. The following data is stored in the table:

* Neighbor node
* Cost to reach other nodes (this could be based on number of hops, or RSSI value from signal strength)

The format of the routing table is:

Destination	Next Hop	Cost
[Dest node ID]	[Next hop node ID]	[Cost]

When the topology changes, the routing table must be updated and shortest path re-calculated. The system initially operates with hop-count cost and it can later be updated to use RSSI values for a more optimal route calculation.

4.6 Data Transmission Process

The SHAM system will support multi-hop data transmission from source to destination:

* The source node creates a data packet with a unique ID for this data transmission.

* The source node checks the routing table to determine the next hop node that is one hop closer to the destination.
* The source node transmits the data packet to the chosen next hop node via Wi-Fi.
* The data packet continues to be transmitted via intermediate nodes until it reaches the destination.
* The destination node acknowledges the reception of the data packet via an ACK message.

4.7 Failure Detection Mechanism

Failure detection in the system will be achieved through timeout monitoring of the neighbors. Each node constantly checks if it receives HELLO messages from its neighbors within a given time limit. If it does not receive a message within the time limit it is considered as having failed, it is then removed from the node's neighbor list and the routing table.

4.8 Self-Healing Mechanism

Upon failure of a node, as stated in 4.7 the routing table will contain obsolete routing information. It becomes imperative for the system to quickly update the routing tables and identify an alternative shortest path to continue reliable communication. This is done automatically by each node, ensuring an almost uninterrupted network flow.

4.9 Message Structure Design

Efficient communication necessitates that messages are small, and the system utilizes three main types of message:

* HELLO Message:
  Used to broadcast a nodes existence in the network to the neighbors.
* DATA Packet:
  Carries information between source and destination nodes.
* ACK Packet:
  Confirms the successful reception of data packets and hello messages.

* ERROR Message:

Conveys failure and link-break messages within the mesh network.

4.10 Working of the Complete System

Each ESP32 node will boot up and initialize Wi-Fi followed by periodic HELLO messages broadcasted to establish network connections and create a neighboring list, thus enabling routing table setup. As data transmission occurs, each node constantly monitors its neighbors and in the case of failure of any of them, rerouting is initiated and the entire process is continued by alternative routes available.

Chapter 5 : Critical Analysis of the Proposed System

SHAM – Self-Healing Autonomous Mesh network protocol SHAM proposes to overcome the limitations of traditional networks through a Self-Healing Autonomous Mesh (SHAM) protocol which integrates dynamic routing, autonomous nodes, and self-healing features in a completely decentralized manner. This section will critically analyze SHAM in terms of performance, reliability, scalability, efficiency, and practical limitations.

5.1 Strengths of the Proposed System

One of the key strengths of the proposed SHAM system is its self-healing capability, which allows the network to automatically recover from node or link failures. Traditional systems in case of failures of a single node or link bring down the entire network or part of it, whereas in SHAM the network dynamically detects dead nodes and recomputes alternate paths for routing, thus ensuring continuity of communications and increasing reliability in critical application areas such as in the case of disasters or during operations for defence forces.

Another advantage of the system is that it is a fully decentralized system, i.e. It is not dependent on a central node, but rather all nodes co-operate to provide communication. This makes the network highly robust and free from a single point of failure. Each node acts as an independent decision making entity to make routes and carry messages for other nodes in the network, increasing the fault tolerance of the network.

The system can dynamically adapt itself to the changes in the network environment. Nodes can join or leave the network at any point of time and routing tables will get updated automatically due to constant neighbour discovery (HELLO messages). This feature enables SHAM to be deployed in mobile or fluid environments.

The system uses multi-hop routing that considerably increases the effective communication range beyond direct wireless capabilities of the nodes. This is a useful feature in large area deployment or when there is obstacle in communication path.

5.2 Performance Evaluation

The proposed SHAM system can achieve efficient data transmission by dynamically adapting routes to minimize the path loss and latency. The routing algorithm can be made such that it finds the shortest path between the source and the destination in terms of hop count (using Dijkstra's algorithm) or minimum latency.

However, the performance of the network is influenced by the number of nodes, network density, and the frequency of control packets (HELLO and routing updates). Small and medium networks will not have very noticeable latency and delay in communication with minimal control overhead, while for large scale networks the control overhead can become significant and decrease efficiency.

The system incurs overhead of HELLO messages and periodically exchanged routing tables for keeping track of neighbours and routes respectively. The frequency of these messages influences network overhead, but is necessary to maintain an up to date network map and avoid outdated routes.

5.3 Scalability Analysis

The decentralised architecture makes the SHAM system highly scalable. Any number of nodes can be added to the network and integrated into it automatically. This feature can enable it for large scale sensor networks and other IoT applications.

However, the scalability of the system is inherently limited by the memory and processing capacity of the ESP32 microcontrollers. Large networks require a significant number of neighbour information, so in very large scale networks, the number of neighbour information can make the memory in each node full.

5.4 Reliability and Fault Tolerance

Reliability is a key strength. The SHAM system's self-healing capability and automatic re-routing ensure high reliability. The use of ACKs (acknowledgement) for packet transmission makes the transmission reliable.

The fault tolerance of the network is also high due to distributed architecture as explained earlier. Even if few critical nodes fail, the network can still continue to provide communication paths (unless critical nodes in a small network fail causing network partitions). Network partitioning in sparse networks where the network is heavily dependent on few links can be an issue.

5.5 Limitations and Challenges

The system may consume considerable power because of continuous exchange of HELLO messages and maintaining a local network map at each node. This limitation makes SHAM not suitable for battery powered systems without proper optimization.

Another limitation is increased latency, since data needs to travel through multiple nodes to reach its destination, as each node introduces some delay.

The implementation of a dynamic and self-healing routing protocol can be challenging as it requires robust algorithms to handle various scenarios like node failures, network joining/leaving, etc.

The decentralised network can be prone to security issues. Since there is no central authority, it is difficult to authenticate nodes, prevent malicious activity, and ensure data security. Proper encryption, authentication and access control mechanisms need to be implemented which can increase complexity.

5.6 Practical Feasibility

The SHAM system, using low-cost ESP32 modules, is practically feasible for many applications, especially where existing wired infrastructure is absent or difficult to deploy, or in highly mobile and unpredictable environments. The low cost and flexibility of ESP32s makes SHAM an affordable and versatile solution.

The practicality of SHAM depends on how well the system is tuned for different environments (indoor/outdoor), density of nodes and nature of mobility. Careful selection of node placement, optimal configuration for power management, and resilience against environmental factors like interference, signal attenuation etc. Are necessary for reliable deployment.

Chapter 6 : Conclusion And Future Scope

6.1 Conclusion

The proposed Self-Healing Autonomous Mesh (SHAM) system presents a viable and reliable mechanism for creating distributed communication networks. By utilizing ESP32 microcontrollers and mesh networking technologies, it successfully eliminates the need for centralized infrastructure and enables independent communication between nodes. Every node in the network has various responsibilities such as neighbor detection, routing, communication, and failure notification.

Self-healing is perhaps the most crucial feature of the SHAM system. Node failure is automatically identified through the timeout mechanism, and the network re-routes data automatically through different paths without any human interference. This feature is highly beneficial in disaster management, military communication and remote sensing applications as it ensures continuous communication without any disruptions.

Furthermore, the dynamic routing protocol ensures optimal communication paths by constantly updating routing tables and dynamically adapting to network topology changes. Multi-hop communication has also been implemented which has a wider reach enabling long-range communication when direct connections are not possible.

Some disadvantages associated with the SHAM system are an increase in power consumption, network overhead, and limitations of the ESP32 device's hardware components. However, the SHAM system provides a robust basis for advanced distributed networks.

This project aims to achieve its objective of design and implementation of a self-healing autonomous mesh network, effectively highlighting how a decentralized system could potentially serve as a flexible and efficient communication network.

6.2 Future scope

The SHAM system has a promising potential to be advanced for future real world application and development. Routing overhead could be reduced, and performance can be enhanced on large and dynamic networks by implementing more advance protocols like Ad hoc On-Demand Distance Vector (AODV) or Hybrid routing techniques.

Another area to focus on is security. Implementing encryption and authentication mechanisms and a safe method for secure key exchange would be critical to ensure system security and prevent unauthorized access. This is important for sensitive application domains such as defense and industrial automation.

Reducing power consumption will also be an important part of future work. Power consumption can be reduced through adaptive HELLO messages, sleep modes, and by implementing energy-efficient communication protocols. This would be ideal for applications involving mobile or battery powered devices.

More development is also possible to support additional sensors and provide connectivity to the cloud enabling real-time monitoring of data collected at the various nodes. The SHAM system would then be capable of applications such as smart cities, environmental monitoring and industrial IoT applications. Implementing machine learning techniques would enable to improve routing by enabling predicting and preventing network problems and therefore optimize routing performance.

Finally large-scale tests in a real-world environment would determine if and how well the system is performing under different circumstances thus identifying further problems to overcome and improving system reliability.

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ESP32 Wi-Fi API Documentation

→ Official documentation explaining ESP32 Wi-Fi features, protocols, and implementation details.

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[2] ESP-IDF (Espressif IoT Development Framework)

ESP-IDF Official Documentation

→ Provides complete development environment and APIs for ESP32-based embedded systems.

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[3] ESP32 Networking APIs and Mesh Support

ESP32 Networking & Mesh APIs

→ Covers ESP-MESH, ESP-NOW, and networking stack used in mesh communication systems.

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[4] ESP32 Datasheet

ESP32 Technical Datasheet

→ Provides hardware specifications, Wi-Fi capabilities, and performance characteristics of ESP32.

---

[5] ESP-MESH Programming Guide

ESP-MESH Programming Guide

→ Explains how ESP32 builds self-organizing mesh networks, including routing, node roles, and automatic reconnection features.

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[6] ESP-WIFI-MESH Programming Guide

ESP-WIFI-MESH Documentation

→ Provides detailed API and architecture for implementing Wi-Fi-based mesh networks on ESP32 devices.

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[7] ESP-BLE-MESH Documentation
ESP-BLE-MESH Guide
→ Describes Bluetooth-based mesh networking, enabling many-to-many communication across large IoT networks.

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[8] ESP-BLE-MESH Overview (Espressif)
ESP-BLE-MESH Overview
→ Highlights features like secure communication, interoperability, and large-scale device networking using Bluetooth mesh.

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[9] Research Paper on ESP32 Mesh Networking Challenges
Practical Challenges of Bluetooth Mesh with ESP32
→ Discusses real-world implementation issues, performance gaps, and improvements in ESP32-based mesh systems.